Why This Language? Historical Symmetry Breaking through Cultural Transmission

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Abstract

In iterated learning of emergent languages, cultural transmission can align semantic category systems and steer learners toward structured, efficient regions of language space. We show that limited transmitted evidence can also break symmetry among multiple structured solutions, biasing independent learners toward a shared, historically contingent reconstruction despite substantial item-level turnover. The target and strength of coordination are empirically distinguishable: developmental provenance can shift what learners reconstruct without necessarily changing how strongly they coordinate. In a controlled referential-game model, anchors captured from an immature snapshot of a parent's language steer descendants toward the immature organization; in the main GRU learner, this occurs without a detectable weakening of coordination under a pre-specified criterion, but at a cost to systematicity. We call this the **Snapshot Effect**. A second architecture and a minimal reconstruction model recover the same symmetry-breaking and provenance-target pattern. The local propagation mechanism is regime-dependent: in the compressed languages of the GRU learner, class-matched anchors preferentially stabilize untaught neighbouring meanings that use the same convention, whereas this mechanism does not carry over to a near-injective learner. Here, symmetry breaking refers to historical evidence resolving underdetermination among multiple viable structured reconstructions, not to competition among labels.

Culture need not create structure. It makes one structured possibility historical.

Keywords: cultural transmission; iterated learning; emergent communication; symmetry breaking; developmental provenance; language evolution.

Plain Language Summary

Two learners exposed to the same parent could each build a good language, but not necessarily the same one. In a small controlled model, we show that a limited set of inherited examples can make independent learners converge on the same organization of meaning even when many individual forms change. The examples also carry information about when they were recorded. If they come from an immature stage of the parent's language, learners are biased toward reconstructing that earlier organization rather than the parent's final one. In the main learner, this shift occurred without a detectable reduction in coordination between descendants. The results suggest that cultural transmission can preserve historical continuity not by copying a language, but by biasing which viable language is reconstructed.

AI Use Disclosure

Generative AI tools (Anthropic Claude via Claude Code and OpenAI Codex) were used to assist with code development, experimental execution, adversarial review, literature discovery, and language editing. The author formulated the research questions, established the pre-registration framework and interpretive constraints, reviewed and verified all analyses and outputs, and takes full responsibility for the manuscript.

Supplementary Material and Code

The accompanying Supplementary Material contains the complete Methods, evidence tables, registered outcomes including failed predictions, and the reconstruction model and move-cost analysis. The accompanying *Version 2 Revision Notes* describe the corrections to the first version. The original preprint and original code, registrations and results remain archived as Version 1; these links identify the earlier release. Instructions for the corrected analyses and their reproducibility limits appear in Supplementary Section S5. Headline numbers are pre-registered unless marked [exploratory]; the tests behind each section are listed at its end and tabulated in the Supplementary Material. All structure claims in the main text use collision-free measures (Supplementary Note 2).

1. Introduction

Two children learn to talk from the same parent. Each of them could have built a language of their own, a different one and just as good. Why do they end up speaking the same one? And if the parent's speech was recorded while the parent was still learning, what exactly do the children inherit: the language, or the moment it was recorded?

Prior work has answered neighbouring questions. Iterated learning through a transmission bottleneck makes emergent languages more structured and more learnable (Kirby, Cornish & Smith 2008; Ren et al. 2020; Li & Bowling 2019); transmission aligns speakers' category systems beyond what shared biases produce (Silvey, Kirby & Smith 2019) and steers them toward efficient regions of the space of category systems (Xu, Dowman & Griffiths 2013; Carr et al. 2020; Carlsson, Dubhashi & Regier 2024). What these accounts leave open is the case where many structured languages are already available. Our claim is not that transmission causes convergence; that is known. The unresolved question is not why learners converge, but what determines the identity of the solution on which they converge: how does a lineage end up speaking one of them rather than another, and what is the thing that is passed on?

We answer with a miniature version of the situation. It is a minimal controlled model built to isolate the causal effects of transmission, not a model of human language. Neural agents play a referential game over 64 objects, every combination of three attributes with four values each, so the "meaning space" is a small grid of things to talk about, and a language is a way of assigning each of them a three-symbol message. Each generation is a fresh pair that first learns 19 of its parent's (object, message) pairs from a record and then trains on the game. Because generation 0 is bit-identical across conditions, the parent, the children and the record can each be held fixed while the others vary: two independently initialised children can be given the same or different records, and the record can be sampled from the parent at different stages of its development.

Three claims follow. **(1) Structured underdetermination.** Learners build structured languages without any record, but do not reliably reconstruct the same one. **(2) Anchored symmetry breaking.** A limited set of transmitted examples biases independent learners toward the same reconstruction: among the alternative structured reconstructions available to them, limited evidence biases them toward one. The symmetry is precise: multiple reconstructions of comparable structure and task value remain viable; shared task and learner biases constrain the space without uniquely selecting one, and the record supplies the additional asymmetry. We use the term differently from the naming-game literature, where symmetry breaking is a population resolving competition among labels (Baronchelli et al. 2006; Puglisi et al. 2008). **(3) Developmental provenance.** The stage at which the evidence was captured biases what is reconstructed; how strongly learners coordinate, and through which local mechanism, depends on the learner and representational regime. A second learner with a different learning rule, and a minimal reconstruction model with no learning at all, recover the same symmetry-breaking and provenance-target pattern, so the pattern is not an artefact of one architecture, though the model demonstrates sufficiency in one setting, not universality. The core confirmatory hypotheses were registered before their confirmatory analyses on data not used for their discovery; failures and later robustness tests are reported in full in the Supplementary Material.

2. Results

2.1 Structured underdetermination and symmetry breaking

Learners can build structure on their own; what they do not reliably do alone is reconstruct the same one. A shared record makes them converge much more strongly.

A fresh sender–receiver pair trained with no record reaches structure of the same order as lineages with one: collision-free geometry within a few hundredths of rewritten-record lineages, while concept matching is lower by 0.05–0.07 (rewrite − no record CBM +0.048 / +0.070 / +0.066 across selection rules, all CIs positive) [exploratory]. What it does not reach is a particular solution. We measure how similarly two languages group meanings into messages with the adjusted Rand index (ARI): the chance-corrected agreement between two partitions of the training objects into same-message classes, 1 for identical grouping and 0 at chance. Because it compares groupings rather than the messages themselves, it separates inheritance of organisation from copying of forms. Two no-record children of the same parent agree at ARI 0.08–0.10, and a no-record child agrees with its parent at 0.06. At the item level the alternatives carry similar communicative value (per-object accuracy 0.92–0.95 whether an object keeps the parent's form or a neighbour's) [exploratory]; similar item-level accuracy is consistent with, but does not establish, functional equivalence of whole alternative languages. The solution space is structured but underdetermined.

Yet a limited transmitted record is enough to break the symmetry. Two independently initialised children taught the same 19 pairs agree at ARI 0.46; taught two different random 19-sets, 0.25; taught nothing, 0.10 (Fig. 1a), and again on twenty untouched seeds (0.43 / 0.26 / 0.05). Siblings with the same record resemble each other (0.46) more than either resembles the parent (0.31; Fig. 1b), and this holds on objects neither sibling was taught (0.34 vs 0.24) [exploratory]: the record constrains reconstruction beyond the items it contains. Lineages inherit the same way: with a rewritten record, parent–child ARI exceeds the no-record baseline by +0.23 to +0.64 across four channel families, and again on twenty untouched replication seeds. All of this coexists with substantial turnover: only 31–55% of training objects keep the parent's exact form across a generation [exploratory]. In the compressed GRU regime, the organisation into classes is inherited more reliably than the forms that realise it.

The symmetry can be stated without metaphor. Call a language *viable* if it reaches the structure and accuracy range observed in this task; the opening of this section shows that many partitions of meaning space are viable at once. The symmetry is a property of the learner given the task alone: it does not distinguish among viable partitions, so independently initialised learners land on different ones (sibling ARI near chance) while matching each other in structure and accuracy. Breaking that symmetry means supplying evidence that does distinguish among them: a record consistent with one partition and not the others. The claim is not that a record makes learners converge in some general sense, which shared biases already do, but that it selects which viable partition they converge on, and that the selection is historical, fixed by what happened to be transmitted rather than by task value. Two observations separate this from mere seed variance: each viable partition is stable enough to be inherited when it is transmitted (the lineage result above), and the alternatives carry similar communicative value at the item level. Whether the task ranks complete alternative languages was not tested directly; a direct test would compare them on held-out communication.

Takeaway: what is transmitted is a bias on reconstruction, not a copy: siblings with the same record resemble each other more than they resemble the parent, even on items neither was taught.

Pre-registered tests in this section: C6, K16, K13a (Supplementary Tables S1–S4).

2.2 Developmental provenance biases the reconstruction target

Give both siblings a record taken early, before the parent had settled: they still agree with each other, but on the parent's early language.

We taught the same 19 objects with the parent's forms sampled either at the end of training (mature) or at step 500 of 2,000 (immature; 72% of these forms differ from the final ones). Crucially, immature anchors steer reconstruction toward the immature organisation: siblings taught the immature snapshot align with the parent's step-500 partition at ARI 0.43 vs 0.21 for mature anchors (Fig. 3). In the GRU learner they met the pre-specified criterion for no material weakening of coordination (mean difference ≥ -0.05 with CI lower bound ≥ -0.10, a one-sided criterion, not a demonstration of equality): sibling ARI 0.55 vs 0.46 (replication +0.08). Alignment with the parent's *final* language is lower with immature anchors (0.28 vs 0.35 on 45 seeds), but only in 73% of seeds, below our rule, reported as a consistent but unconfirmed effect. The reconstruction model illustrates one reason this need not occur: where the early and final partitions of a parent overlap (ARI 0.70 in the model), alignment with the early state need not reduce alignment with the final one. At the item level, a child taught an object's mature form ends with the parent's final form 79% of the time; taught the step-500 form, 26% (Fig. 4d; per-seed means over 65 seeds): learners reconstruct toward the organisation they were given rather than automatically recovering the mature one. This is the Snapshot Effect: transmitted evidence biases reconstruction toward the developmental state in which it was captured.

A record that accumulates forms carved during the parent's training tends to preserve an early developmental state, and lineages transmitting it build less systematic languages than lineages whose record is rewritten from the parent's final language (Fig. 4a). The child must replace more of what it was taught (fidelity to the record falls to 0.66 within a generation vs 0.82 for mature anchors; Fig. 4b), and excluding every object that carries a stale form closes only ~15–20% of the gap [exploratory], suggesting that much of the cost is systemic. Under "hard" selection, where record slots turn over every generation, the advantage appears in only 57–80% of seeds, consistent with forms having less time to become stale.

Takeaway: what learners coordinate on and how strongly they coordinate are distinguishable quantities; the developmental provenance of transmitted evidence biases the first, and its effect on the second depends on the learner (§2.4).

Pre-registered tests in this section: K17c, K17a, K17b, H3, C4 (Supplementary Tables S1–S4).

2.3 How the bias propagates in compressed languages

A child sees only a limited set of inherited examples, yet an inherited example can stabilise nearby meanings the child never saw, especially when they use the same convention. We call such an example a class-matched anchor.

In the compressed languages of the GRU learner (24 of 64 messages distinct; 75% of training objects have a neighbour with the same form), the taught items are learned by supervised imitation, and their influence spills over to neighbours that were never taught. An untaught object keeps the parent's form in 0.46–0.60 of cases when a taught Hamming-1 neighbour shares that form, and in 0.09–0.26 when its taught neighbours carry other forms, close to the 0.10 seen with no taught neighbour at all (gap +0.30 to +0.44 across four channel families each; replication +0.21 to +0.43; Fig. 2a). Retention rises with the number of taught neighbours, and the anchor transmits a *choice*: when the parent had borrowed a neighbour's form for an untaught object, the child repeats that choice in 0.56–0.61 of cases if the neighbour was taught and at chance (0.16–0.20) if not (Fig. 2b). Consistent with a mechanism that works through anchors rather than item identity, four non-random rules for choosing which 19 items to transmit differed from a random choice by less than the pre-set point-estimate bands in the pooled 45-seed cohort (three of four in replication, where the whole-classes rule fell outside the band at −0.031; two of four in the original 15-seed test). The record manipulation is causal, but the same-form versus other-form split within it is a conditional comparison and does not by itself rule out object-difficulty effects. The record is partial rather than exhaustive: 19 of 48 training meanings (30% of the full 64-object world), enough to leave most untaught objects with at least one anchor. Capacity is partly associated with coverage: a record of 8 items leaves most objects unanchored and yields the least structured languages (−0.07), founder intelligibility rises by +0.20 per capacity step, and at equal anchor counts retention was equivalent across capacities only for well-anchored objects (three or more taught neighbours, an additional subgroup); for the registered subgroup of two or more taught neighbours it was equivalent in confirmation but not in replication, and for objects with no anchor it failed twice (Fig. 2c).

Takeaway: in compressed languages, a transmitted example affects nearby meanings that were never taught; which examples are transmitted matters much less than whether they anchor their neighbourhoods.

Pre-registered tests in this section: K15, K11, K10, K14, K1, K4, K12 (Supplementary Tables S1–S4).

2.4 Boundaries and generality

Three ways to break the story, two ways to test whether it is about our learner, and one model with no learning at all.

Persistence. With mature anchors the founder's partition is still detectable five generations later (ARI 0.17); with immature anchors, repeatedly transmitted, it is substantially weaker. History persists across the generations observed, but weakens. After a *single* immature snapshot, redirection weakened over the next two refreshed generations, but persistence was inconclusive (S), so the persistence claim is scoped to recurring immature provenance.

Regime. Under very low exploration, where the parent's language changes little after the snapshot (26% of accumulated entries stale vs 35%), the Snapshot Effect weakened and missed the seed rule, although its mean stayed positive (+0.05); with generations three times longer, where the parent moves further, it held (+0.05; staleness 53%). Developmental divergence after capture, rather than snapshot age as such, appears to moderate the effect.

Making the task more decisive (Hamming-1 distractors in 0 / 50 / 100% of rounds) lowers the partition-level sibling gap in the predicted direction (0.26 → 0.14 → 0.09) but not decisively, while the form-level gap remains within the pre-set equivalence band: that cultural leverage is highest under structured underdetermination remains a prediction with trend-level support.

Scale and coverage. With very low coverage (capacity 8) the maturity advantage in partition inheritance disappears or reverses (−0.10 at capacity 8 vs +0.37 to +0.39 at capacity 40). In a world four times larger with hard distractors (256 objects, half of all rounds requiring more than one attribute), symmetry breaking and partition inheritance hold (+0.10, +0.16 and +0.29), class-matched anchoring keeps its direction but not its magnitude (+0.07; band set at 0.15), and the structural cost of immature records is inconclusive at n = 10 despite staler records (56%); anchor leverage, not staleness alone, appears to limit how much redirection propagates.

Learner and model. An MLP sender and receiver trained end-to-end with straight-through Gumbel-softmax (no REINFORCE, no exploration bonus) produce nearly injective languages (59 of 64 messages distinct; 4% of objects have a same-form neighbour), an empirical property of this learner. Partition measures are uninformative there (two fully injective languages have identical singleton partitions), so we registered form-level tests on fresh seeds. Partition-level tests in this learner were re-scored after a metric correction (identical singleton partitions now score 1): partition inheritance with a rewritten record (A1) is supported in all four channel families under the corrected metric (Version 1.0 reported it as partial), while the sibling-coordination contrast (A4) no longer meets its band; the form-level conclusions are unchanged (Supplementary Table S4). Symmetry breaking holds: sibling form agreement 0.44 with the same record, 0.21 with different records, 0.00 with none (Fig. 1c). Immature anchors redirect the target here too (+0.09) but coordinate siblings somewhat *less* well (0.37 vs 0.44), so the "without detectable weakening" finding belongs to the GRU regime. Class-matched anchoring vanishes except at the smallest capacity. This is a regime boundary of the class-matched mechanism, not a failure of the claim: the broader symmetry-breaking effect persists. We interpret this as a consequence of there being almost no message classes to anchor, so the mechanism belongs to categorical, compressed languages (a hypothesis, not a demonstrated geometric account). Anchoring gradients and anchored choice remain.

The cleanest demonstration that the pattern does not require neural learning comes from a model with no learning at all. A search that minimises collisions + number of classes + cut Hamming-1 edges greedily, with anchors as nodes whose labels are fixed, reproduces the pattern: same anchors > different > none (ARI 0.41 > 0.27 > 0.24); the class-matched spillover emerges without being built in (0.47 vs 0.32 vs 0.31); early-partition anchors bias reconstruction toward the early partition. The persistence comparison (T4) was found on re-inspection not to have implemented the registered frozen-record condition; implemented as registered, it does not reproduce the neural persistence pattern (Supplementary S3, Table S4). The symmetry-breaking, class-matched and provenance patterns can arise without neural learning, from underdetermined structured reconstruction under sparse historical constraints (a measured move-cost analysis replaces the earlier derivation; Supplementary S3). This is a sufficiency result in one setting, not a demonstration of universality.

Takeaway: mechanism regime-specific, principle robust across the settings tested. What carries across is not a particular form of inheritance but the computational role of historical evidence: limited constraints break symmetry among reconstructable solutions and their developmental provenance biases the target; how strongly they coordinate, and through which local mechanism, depends on the learner and representational regime.

Pre-registered tests in this section: K18, E1, L1, D1, D2, K13b, M1–M4, A3′, A4′ (Supplementary Tables S1–S4).

3. Discussion

In one sentence: learners can build a structured language alone, but a limited set of inherited examples biases which one they build, and the examples carry the developmental state in which they were captured.

The role of cultural transmission depends on the regime of the learning problem. When structure is not reconstructable without transmission, the bottleneck creates it (Kirby et al. 2008; Ren et al. 2020); when structure is already reconstructable but underdetermined, the regime studied here, transmission acts primarily as historical symmetry breaking. Prior work treated languages as partitions of meaning space and compared them with the adjusted Rand index (Perfors & Navarro 2014; Xu et al. 2013); our use of the measure is to separate inherited organisation from item-level copying and to compare independently reconstructed descendants. This suggests a mechanistic interpretation of the alignment reported by Silvey et al. (2019): transmission coordinates semantic systems not by copying them but by resolving underdetermination during reconstruction. It also sits between cultural attraction theory, which holds that transmission is reconstructive (Claidière et al. 2014; Scott-Phillips 2017; Acerbi 2021), and iterated-learning theory, which holds that transmitted data are asymptotically irrelevant (Griffiths & Kalish 2007): shared biases explain why languages occupy a structured region of solution space, and limited historical evidence biases where within that region independent learners converge, a finite-time historical contingency compatible with asymptotic attraction. Carlsson et al. (2024) showed that many colour-naming systems are comparably efficient; we show that a small inherited record biases fresh learners toward particular alternatives.

That the developmental stage of a teacher matters is not new: intermediate checkpoints can be better distillation teachers than converged ones (Cho & Hariharan 2019; Wang et al. 2022; Panigrahi et al. 2025), and seeded iterated learning varies how long a teacher interacts before it is imitated (Lu et al. 2020). Those results ask which stage teaches better; ours asks whether stage changes what is reconstructed. We are not aware of prior work that isolates the dissociation between coordination target and coordination strength under a stage manipulation with items held fixed. Nor is this a claim that learners cannot regularise imperfect input: children do (Singleton &

Newport 2004; Hudson Kam & Newport 2005, 2009), Nicaraguan Sign Language cohorts added structure (Senghas et al. 2004), and zebra-finch song returns to wild type over generations (Fehér et al. 2009). It is a claim about identifiability under historical constraints: when a limited anchor set provides sufficient coverage in this reconstruction regime, learners coordinate around the organisation encoded in the captured evidence rather than inferring the source's later organisation.

The three forces this separates are often conflated: inductive and communicative biases shape the space of viable structured solutions; cultural transmission breaks symmetry among those solutions, making one historically contingent organisation recurrent; and the developmental provenance of transmitted evidence biases reconstruction toward one organisation (in the GRU learner without a detectable weakening under the pre-specified criterion, in a second learner with some loss), while repeated transmission of that provenance shapes how long its historical influence persists.

Limitations. A small world, small learners, and predictions that did not hold: the registered capacity–coverage equivalence did not replicate for objects with two or more anchors and failed for unanchored objects; the reconstruction model's persistence comparison did not reproduce the neural result once implemented as registered; under very low coverage persistence can beat freshness; convexity is not a product of accumulation; the reduced alignment of immature-anchored children with the parent's final language reached 70–73% of seeds; the class-matched gap shrinks in a larger world and vanishes in a near-injective learner; and the degeneracy scan gave a trend, not a verdict. Ownership of shared messages is a mechanical consequence of the receiver's independent, deterministic candidate scoring; its consequences are not. All pre-registered outcomes, including these, are tabulated in Supplementary Tables S1–S4.

The question in our title therefore has a historical answer once functional constraints leave multiple viable solutions: this language, because these were the examples that happened to be transmitted, captured at this stage of the parent's development. Culture does not have to preserve particular solutions. It can transmit limited historical evidence that biases independent learners toward the same historically contingent region of solution space, and that evidence encodes the developmental state in which it was captured.

Culture need not create structure. It makes one structured possibility historical.

4. Methods

Essentials only; the full specification is in Supplementary Methods. Objects are the 64 combinations of 3 attributes × 4 values; 16 are held out (never targets, but present as distractors). A round: the sender emits 3 symbols from a vocabulary of 8; the receiver picks the target among 5 candidates. GRU sender and receiver (hidden 64); REINFORCE with an entropy bonus of 0.02 for the sender, cross-entropy for the receiver; 2,000 steps of 64 rounds per generation; six generations; every generation a fresh pair. Before training, the child sender is taught up to *capacity* (8/19/40) recorded pairs by 200 supervised steps. The record's freshness is either *accumulate* (forms carved during the parent's training under an incumbent rule, so most entries are fixed early) or *rewrite* (the parent's final forms). Measures: topsim and its collision-free variant, concept-best matching, ownership (the receiver's decode of each message among all 64 objects), and ARI between languages' partitions of the 48 training objects. Anchors are taught Hamming-1 neighbours of an untaught object; class-matched anchors share its parent form; provenance is the parent step at which the recorded form was sampled. Sibling design: two children with different initialisations taught the same 19 pairs, different 19-sets, or nothing. Snapshot design: the same 19 objects with step-500 vs final forms. Second architecture: MLP agents with straight-through Gumbel-softmax, no REINFORCE, no entropy bonus. Medium world: $4 \times 4 = 256$ objects, message length 4, capacity 77, Hamming-1 distractors in half of all rounds. Decision rule for every registered prediction: $\geq$ 80% of seeds in the predicted direction and a paired bootstrap 95% CI excluding 0; equivalence-type claims state their band in advance (the K17a criterion is one-sided: no material weakening, not a symmetric equivalence test). All uncertainty statements and error bars use the seed as the independent unit. The repository history available cannot independently date each threshold; the registration files document the stated chronology. Registered-before-run and registered-before-computation hypotheses are distinguished in the registration files; the replication pipeline was run by one command on seeds 100–119; K16 and K17a/c were added to the driver after that run and evaluated from the same saved parents, and K18 was added to the registration while the replication was running, before inspection. Replication on arbitrary fresh seeds has not been verified end-to-end (Supplementary Methods, Compute).

Figures

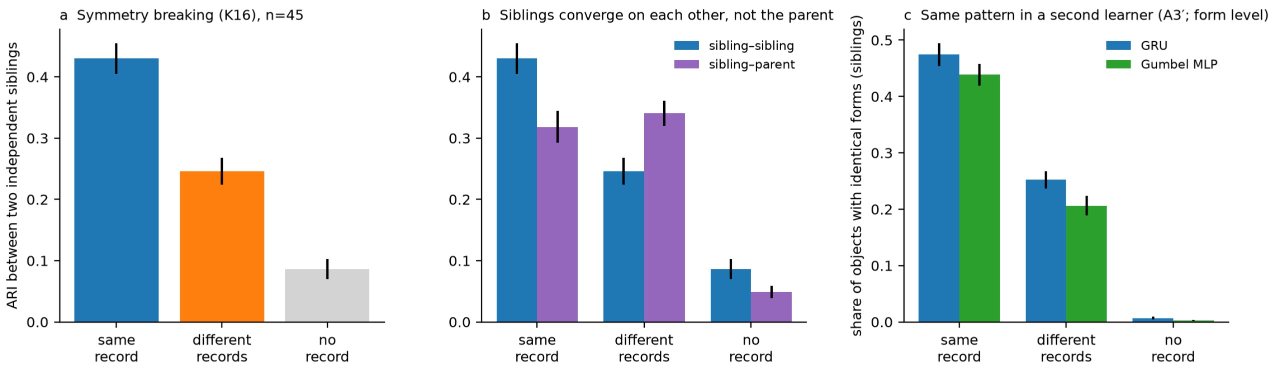

Figure 1. Historical symmetry breaking among independently trained siblings. Two children of the same parent, trained separately, become substantially more similar when they were shown the same limited set of examples: same record → siblings look alike; different records → less alike; no record → almost unrelated. (a) Partition similarity (adjusted Rand index over the 48 training objects) between two independently initialised siblings taught the same 19 (object, message) pairs, two different random 19-sets, or nothing; n = 45 seeds (30 discovery + 15 pre-registered, K16). (b) Siblings taught the same record resemble each other more than either resembles the parent. (c) The same pattern in a second learner: an MLP sender and receiver trained with straight-through Gumbel-softmax (no REINFORCE, no exploration bonus) produce nearly injective languages, so agreement is measured as the share of objects on which the two siblings use identical forms; GRU (seeds 0–29) vs Gumbel MLP (seeds 120–159; A3′, pre-registered). Error bars: s.e.m. over seeds.

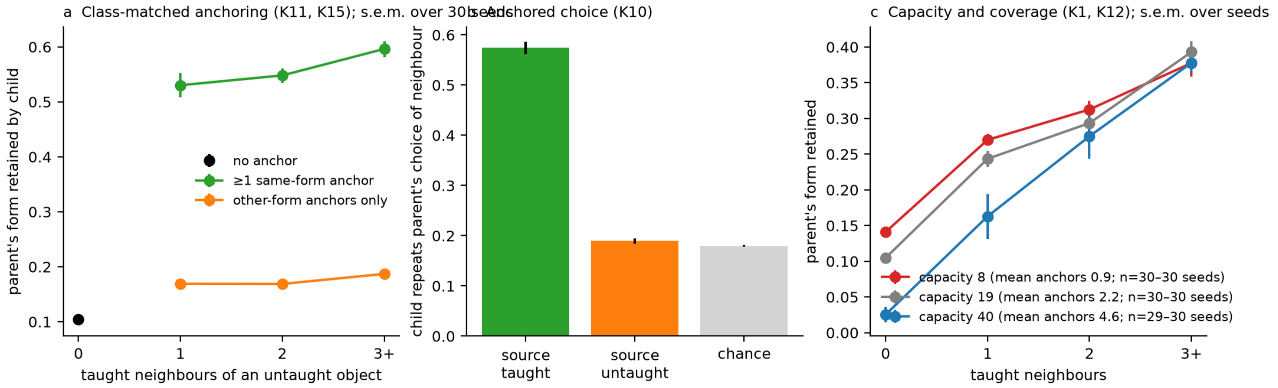

Figure 2. Class-matched anchors propagate transmitted conventions. A transmitted example steadies the meanings around it, especially the ones that use its convention. (a) Share of untaught objects that keep the parent's form, by number of taught neighbours, split by whether at least one neighbour shares the object's own form (K11, K15). (b) When the parent had borrowed a neighbour's form for an untaught object, the child repeats that choice if the neighbour was taught, and at chance if not (K10). (c) Part of the benefit of a bigger record is more anchors per object: at equal anchor counts, retention is similar across capacities for well-anchored objects (three or more taught neighbours, an additional subgroup); the registered subgroup of two or more did not replicate equivalence, and unanchored objects retain less at capacity 40 (K1, K12). Error bars in all panels: s.e.m. over seeds (n = 30 in every stratum except capacity 40 with one anchor in panel c, n = 29: one seed had no eligible observation). Chance in panel b is computed on the same untaught-source cases as the bar it is compared with.

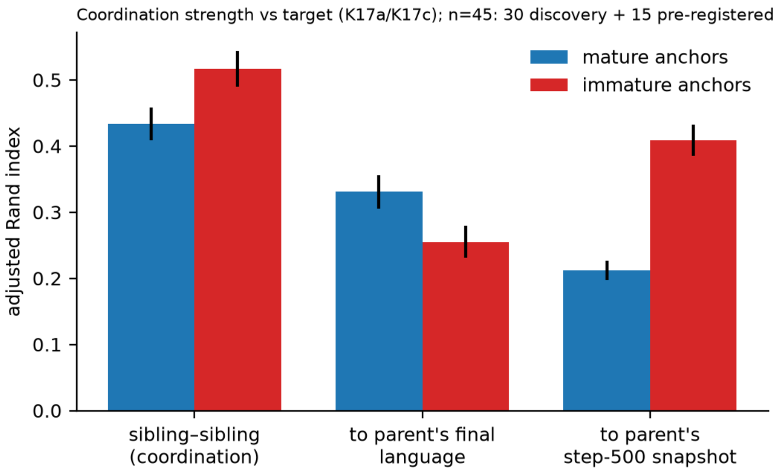

Figure 3. Developmental provenance shifts the reconstruction target. In the GRU learner, examples taken from an immature stage of the parent's language coordinate children without a detectable weakening under the pre-specified criterion, but toward the immature version (in the second learner they coordinate somewhat less well; §2.4). Bars: sibling–sibling similarity (coordination), similarity to the parent's final language, and similarity to the parent's step-500 snapshot, for children taught mature vs immature forms of the same 19 objects (K17a, K17c; n = 45: 30 discovery + 15 pre-registered).

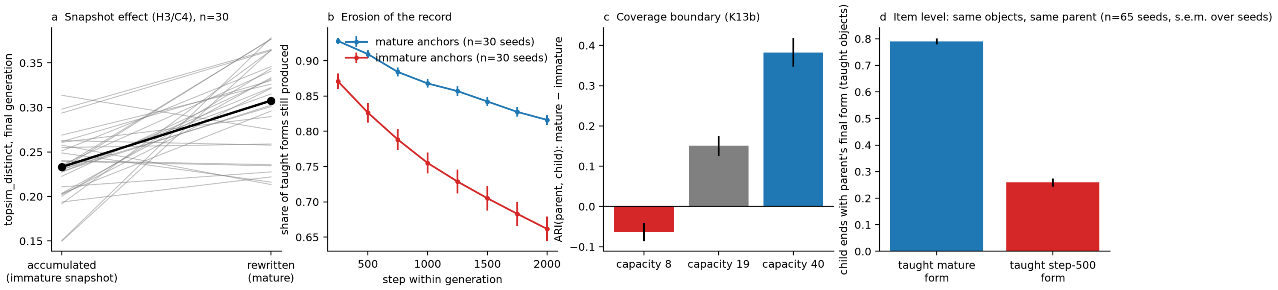

Figure 4. The Snapshot Effect and its structural consequences. (a) Lineages whose record accumulates early forms build less structured languages than lineages whose record is rewritten from the parent's final language, paired by seed (H3, C4; collision-free geometry; on this panel's population, all 64 objects at the final generation, rewrite − accumulate = +0.074 in 24 of 30 seeds). (b) Within a generation, communication erodes what imitation installed, twice as fast when the record is immature. (c) With a very small record (capacity 8) the advantage of mature records disappears or reverses; with a large one it is strong (K13b). (d) Same objects, same parent: a child taught the mature form ends with the parent's final form 79% of the time; taught the step-500 form, 26% (per-seed means, n = 65 seeds: 30 discovery, 15 pre-registered, 20 replication). Error bars in all panels: s.e.m. over seeds.

References

Acerbi, A. (2021). Culture without copying or selection. Evolutionary Human Sciences, 3, e50.

Baronchelli, A., Felici, M., Loreto, V., Caglioti, E., & Steels, L. (2006). Sharp transition towards shared vocabularies in multi-agent systems. J. Stat. Mech., P06014.

Carlsson, E., Dubhashi, D., & Regier, T. (2024). Cultural evolution via iterated learning and communication explains efficient color naming systems. J. Language Evolution, 9, 49–66.

Carr, J. W., Smith, K., Culbertson, J., & Kirby, S. (2020). Simplicity and informativeness in semantic category systems. Cognition, 202, 104289.

Cho, J. H., & Hariharan, B. (2019). On the efficacy of knowledge distillation. ICCV.

Claidière, N., Scott-Phillips, T. C., & Sperber, D. (2014). How Darwinian is cultural evolution? Phil. Trans. R. Soc. B, 369, 20130368.

Fehér, O., Wang, H., Saar, S., Mitra, P. P., & Tchernichovski, O. (2009). De novo establishment of wild-type song culture in the zebra finch. Nature, 459, 564–568.

Griffiths, T. L., & Kalish, M. L. (2007). Language evolution by iterated learning with Bayesian agents. Cognitive Science, 31, 441–480.

Hudson Kam, C. L., & Newport, E. L. (2005). Regularizing unpredictable variation. Language Learning and Development, 1, 151–195.

Hudson Kam, C. L., & Newport, E. L. (2009). Getting it right by getting it wrong. Cognitive Psychology, 59, 30–65.

Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory. PNAS, 105, 10681–10686.

Li, F., & Bowling, M. (2019). Ease-of-teaching and language structure from emergent communication. NeurIPS.

Lu, Y., Singhal, S., Strub, F., Courville, A., & Pietquin, O. (2020). Countering language drift with seeded iterated learning. ICML.

Panigrahi, A., Liu, B., Malladi, S., Risteski, A., & Goel, S. (2025). Progressive distillation induces an implicit curriculum. ICLR.

Perfors, A., & Navarro, D. J. (2014). Language evolution can be shaped by the structure of the world. Cognitive Science, 38, 775–793.

Puglisi, A., Baronchelli, A., & Loreto, V. (2008). Cultural route to the emergence of linguistic categories. PNAS, 105, 7936–7940.

Ren, Y., Guo, S., Labeau, M., Cohen, S. B., & Kirby, S. (2020). Compositional languages emerge in a neural iterated learning model. ICLR.

Scott-Phillips, T. C. (2017). A (simple) experimental demonstration that cultural evolution is not replicative, but reconstructive. J. Cognition and Culture, 17, 1–11.

Senghas, A., Kita, S., & Özyürek, A. (2004). Children creating core properties of language. Science, 305, 1779–1782.

Silvey, C., Kirby, S., & Smith, K. (2019). Communication increases category structure and alignment only when combined with cultural transmission. J. Memory and Language, 109, 104051.

Singleton, J. L., & Newport, E. L. (2004). When learners surpass their models. Language, 80, 370–407.

Wang, C., Yang, Q., Huang, R., Song, S., & Huang, G. (2022). Efficient knowledge distillation from model checkpoints. NeurIPS.

Xu, J., Dowman, M., & Griffiths, T. L. (2013). Cultural transmission results in convergence towards colour term universals. Proc. R. Soc. B, 280, 20123073.

Supplementary Material: Why This Language? Historical Symmetry Breaking through Cultural Transmission

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Previous version (1.0): archived preprint
Original code and results: Version 1 archive

Companion to the main article. See also the accompanying Version 2 Revision Notes. Contents: S1 Supplementary Methods; S2 Supplementary Tables S1–S4 (every pre-registered outcome, including failures); S3 the minimal reconstruction model and its move-cost analysis; S4 Supplementary Figure Legends; S5 registration timeline and reproduction; S6 references. Test codes (K16, A3′, ...) are the identifiers used in the registration files `results_*/PREREG.md` and in `manifest_k.json`. Effects are paired differences by seed with 95% bootstrap CIs (5,000 resamples); "k/n" is the number of seeds in the predicted direction. The decision rule, fixed in advance, is ≥ 80% of seeds in the predicted direction AND a CI excluding 0 (directional), or a pre-stated band (equivalence). Verdicts follow that rule mechanically; where a result missed it but had a CI excluding 0, it is labelled "CI > 0, seed rule not met".

S1. Supplementary Methods

Environment and task

Objects are the full product of 3 attributes × 4 values (64 objects), encoded one-hot (12 dimensions). For each seed, 25% of the objects (16) are held out: they are never targets during training or teaching, but they do appear as distractors (distractors are drawn from all other 63 objects), so the receiver sees held-out objects as candidates without ever being rewarded for picking them. A round is a Lewis referential game: the sender sees a target and emits a message of 3 symbols from a vocabulary of 8 (512 possible messages); the receiver sees the message and 5 candidates (the target plus 4 distractors drawn without replacement, uniformly from the other 63 objects) and picks one. Reward is 1 for a correct pick. Distractor sampling is a bijection (no duplicate distractors; the earlier duplicate bug affected 0.27% of rows and was fixed before all v3 runs).

Agents

Sender: linear encoder (12 → 64, tanh), a GRU cell (hidden 64) that emits the 3 symbols autoregressively from a start token, with a linear output head; symbols are sampled during training and taken greedily for all reported languages. Receiver: the message is embedded and read by a GRU (hidden 64); each candidate is scored independently by a dot product between the message representation and a linear projection of the candidate's one-hot encoding; the candidate with the highest score is chosen. Because scoring is independent and deterministic, two objects that share a message always resolve to the same candidate (the "owner"); this mechanical fact defines ownership and orphans (Supplementary Note 1).

Training

Each generation is a fresh sender–receiver pair (new initialisation), trained for 2,000 steps on batches of 64 rounds over the training objects. The receiver is trained by cross-entropy; the sender by REINFORCE with a moving-average baseline (0.95) and an entropy bonus of 0.02 (the boundary experiment E varies this to 0.005 and 0.08). Adam, learning rate 1e-3. Six generations per lineage. Random streams: the world split, the training batches, and the record each have their own seeded generators, so generation 0 is bit-identical across all transmission conditions of a seed, and every cross-condition comparison is paired by seed.

Transmission channel ("record")

Between generations, the child sender is taught up to *capacity* (8 / 19 / 40) (object, message) pairs by 200 supervised steps (learning rate 3e-3) before its own training starts; optionally the child receiver is also taught to pick the recorded object among distractors ("reader = both"). Six factors define the channel: **selection** of which objects occupy record slots (random; by communicative success; "hard" = lowest per-object accuracy), **slot dynamics** (fixed for the lineage; redrawn each generation; dynamic = re-selected each generation), **freshness**, which takes two values. *Accumulate*: during the parent's training, every *successful* round (object o, message m) is offered to the record. If o is not recorded and a slot is free, (o, m) is stored with a success count of 1. If o is recorded with the same m, its count is incremented. If o is recorded with a different m, the incumbent is replaced only if its count is below 2 (a single success); otherwise the new form is discarded. If o is not recorded and the record is full, the entry with the lowest count is evicted only if that count is below 2; ties among counts are broken by dictionary order (first minimum), never by recency. Counts grow into the thousands, so an entry that succeeded twice is in practice permanent. At the end of a generation empty slots are filled with the parent's final greedy forms. *rewrite*: every slot is overwritten with the parent's final greedy form at the end of the generation (count reset to 1). The remaining factors are **capacity**, **noise** (each recorded symbol independently corrupted with probability 0 / 0.2), and **reader** (sender / both). The named v2 conditions (pair, population, generations, oral, oral_fixed, bone, bone_edition) are cells of this grid. A "generations" cell has no record (fresh pair every generation); "pair" trains a single pair for 12,000 steps.

Logging

Every 250 steps: the sender's full greedy language (64 messages), the receiver's decode map (which of the 64 objects each of the 64 messages resolves to), message entropy, train / held-out accuracy. At every generation end: per-object accuracy, record contents, slot set, transmitted objects, and both agents' weights.

Measures

- **topsim**: Spearman correlation (tie-corrected) between attribute Hamming distances and message Hamming distances over all pairs of the 64 objects (held-out included; the sender produces a greedy message for every object); reported for comparability but inflated by semantically local message collisions (Supplementary Note 2).
- **topsim_distinct**: the same over pairs with distinct messages only (collision-free geometry).
- **CBM** (concept–best matching): over all 64 objects, a 24 × 12 co-occurrence matrix between (position, symbol) and (attribute, value) is built from the greedy language; a one-to-one Hungarian assignment maximises total co-occurrence, and CBM is that total divided by 64 × 3 (the number of symbol slots), so CBM = 1 iff every symbol slot is explained by one (attribute, value) under a consistent assignment. Sibling and content-rule comparisons additionally report topsim_distinct and CBM restricted to the 48 training objects where stated.
- **convexity**: share of message classes (≥ 3 objects) that are connected in the Hamming-1 graph of objects.
- **decode map / ownership**: for each of the sender's 64 greedy messages, the receiver scores all 64 objects as candidates and the arg-max is the decoded object (floating-point arg-max,

deterministic). Object o *owns* its message if the decode of its own message is o; otherwise o is an *orphan* and the decoded object is its owner.

- **ARI**: adjusted Rand index (Hubert & Arabie) between two languages' partitions of the 48 training objects into same-message classes; used for parent–child, sibling–sibling, and generation-gap comparisons. Held-out objects are excluded because their forms are never reinforced.
- **continuity**: the last generation's receiver plays 512 rounds with the generation-0 sender's greedy messages: targets are training objects, candidates are the target plus 4 distractors drawn from all other 63 objects (held-out included), with a fixed evaluation seed; continuity is the accuracy. Chance is 0.20.
- **anchor**: a taught training object that is a Hamming-1 (one attribute differs) training neighbour of an untaught training object. **Class-matched (same-form) anchor**: an anchor whose recorded form equals the untaught object's own parent form, i.e. both belong to the same message class in the parent. **Provenance** of an anchor is the parent training step at which its recorded form was sampled: *mature* = the parent's final (step-2,000) form; *immature* = the step-500 form (snapshot design) or, in accumulate cells, the carved form, which typically differs from the parent's final form. Anchor counts in K10–K12 and K15 use mature anchors only (recorded form == parent's final form); the snapshot design (K17) varies provenance while holding the anchor set fixed.

Experiments

- Factorial sweep: 77 cells × 10 seeds (`results_v3`); confirmation 1: 10 cells × 20 new seeds (`results_v3_confirm`); confirmation 2: 24 cells × 20 seeds (`results_v3_confirm2`); content-rule, sibling and snapshot experiments on seeds 30–74 (parents trained fresh); independent replication on seeds 100–119 by one command (`replicate.py`; K16 and K17a/c were added to the driver after that run and evaluated from its raw files, the K16 sibling arms being trained afterwards from the same saved parents).
- **Sibling design**: two children with different initialisations and training streams are taught from the same parent (end of generation 0): the same 19 random training objects with the parent's final forms; two different random 19-sets; or nothing.
- **Snapshot design**: the same 19 objects taught with the parent's forms at step 500 (immature; 72% differ from the final forms) vs at step 2,000 (mature).
- **Content rules**: 19 taught objects chosen at random / owners only / orphans only / whole homonym classes / the 19 most stable.

Medium world (boundary experiment M)

4 attributes × 4 values = 256 objects (64 held out), vocabulary 8, message length 4, 4,000 steps per generation, capacity 77 (≈ 40% of the 192 training objects, the analogue of 19/48). Half of all rounds (hard_frac = 0.5) draw their 4 distractors from the target's Hamming-1 neighbours, so a single attribute cannot identify the target; the remaining rounds are standard. Cells: the six record cells at capacity 77 and the no-record generation; seeds 0–9; hypotheses M1–M4 registered before running (`results_medium/PREREG.md`). Siblings for M2 are trained as in the main design with 77 taught pairs.

Second architecture (experiment A)

Sender: a two-layer MLP (12 → 64 → 64, tanh) with a linear head producing 3 × 8 logits, one softmax per message position (no recurrence); messages are sampled with straight-through Gumbel-softmax (temperature 1) during training and taken greedily for all reported languages. Receiver: the one-hot message (24 dimensions) is mapped by a two-layer MLP (24 → 64 → 64, tanh) to a message representation, and each candidate is scored by a dot product with a linear projection of its one-hot encoding, as in the GRU receiver. Both agents are trained end-to-end by the receiver's cross-entropy alone (Adam, 1e-3, batch 64, 2,000 steps); there is no REINFORCE term and no entropy bonus. Teaching (200 supervised steps on the record) and all channel factors are unchanged. The learner produces nearly injective languages (59 of 64 messages distinct on average; 4% of training objects have a same-form neighbour vs 75% for the GRU); because partition measures are uninformative in this regime (two fully injective languages have identical singleton partitions and hence ARI = 1 regardless of their forms; the calculations in Version 1.0 assigned 0 to this edge case, corrected on 2026-09-05), the pre-registered ARI tests (A1–A4) were supplemented, after this was observed and before new seeds were run, by form-level tests (A3′, A4′: exact-form agreement between siblings and with the parent's step-500 forms) on seeds 120–159. Registration and decision tree: `results_arch/PREREG.md`.

Degeneracy scan (boundary experiment D)

Same GRU learner and 64-object world, seeds 0–19, with hard_frac = 0 / 0.5 / 1.0 (the share of rounds whose 4 distractors are Hamming-1 neighbours of the target); a fresh generation-0 parent is trained under each hard_frac, and the sibling design (same 19 / different 19 / none) is run once per parent. Measures: the partition-level sibling gap (ARI, same − different) and the form-level gap (exact-form agreement, same − different). D1 (partition gap decreases monotonically with hard_frac, ≥80% of seeds per adjacent pair) and D2 (form gap within a pre-set equivalence band) were registered before running (`results_degeneracy/PREREG.md`). Correction (2026-09-05): D manipulates distractor geometry; it does not control the number of viable partitions. The registration's premise that Hamming-1 distractors force near-injective languages is false: the code (attribute1 + attribute2 + attribute3) mod 4 separates every Hamming-1 pair with four labels (a statement about the task, not about what this learner can learn). D1 remains inconclusive under the corrected metric.

Pre-registration and decision rule

Two kinds of pre-registration are distinguished in the registration files. *Registered before the runs*: H1–H9 (factorial sweep), C1–C6 (confirmation 1), K1–K9 (confirmation 2), K14, K16 and K17 (new seeds), and E1–E4. K18 was added to the registration file while the replication (seeds 100–119) was already running and before anything from it had been inspected (addendum of 2026-09-04). *Registered before computation on held-back data*: K10–K13 and K15, quantities discovered post hoc on seeds 0–29 of the cap-19 cells and then fixed in writing before being computed on the confirmation-2 cells (capacity 8/40, noise, reader), which had not been used for those quantities; the registration notes what had already been inspected on that sweep (its K1–K9 metrics). All registrations are in `results_*/PREREG.md` and the thresholds are frozen in `manifest_k.json`. A directional hypothesis is supported if ≥ 80% of seeds fall in the predicted direction and the paired bootstrap 95% CI (5,000 resamples) excludes 0; equivalence hypotheses state their band in advance. Results that miss the seed rule but have a CI excluding 0 are reported in full and labelled as such. The K17a criterion (mean ≥ −0.05 and CI lower bound ≥ −0.10) is one-sided, a no-material-weakening criterion rather than a symmetric equivalence test, although the registration file calls it "equivalence". K14 uses point-estimate bands with no interval condition; K12 requires both |mean| < 0.05 and a CI inside ±0.08. All uncertainty statements use the seed as the independent unit (corrected on 2026-09-05 for K8 and the figure error bars). The repository history available at the verification of 2026-09-05 has three commits and cannot independently date each threshold; the registration files document the stated chronology. The theory was frozen on 2026-09-04; later experiments are robustness and boundary tests only.

Compute

CPU only (10-core laptop); a 6-generation run takes ~2 minutes; 10 parallel workers. All code, registrations and results are in the repository; `replicate.py --seeds A..B` runs the sweep, the child, sibling and snapshot experiments and the evaluators K1–K18. The tested workflow is seeds 100–119 as actually run (K16 and K17a/c were added to the driver after that run and evaluated there from the same saved parents). The driver still depends on the original K14/K17 raw files and on hardcoded seed and world ranges, so replication on an arbitrary fresh seed range has not been verified end-to-end (verification of 2026-09-05).

Supplementary Note 1 (ownership). Because the receiver scores candidates independently and deterministically, every message resolves to exactly one object over the full 64-object candidate set. An object *owns* its message if its own message decodes to itself; otherwise it is an *orphan* of the decoded object, its *owner*. Ownership is therefore a mechanical consequence of the receiver architecture; the consequences reported in the main text (borrowing of a neighbour's form, elimination of orphans, class-level inheritance) are not.

Supplementary Note 2 (topsim inflation). Plain topographic similarity counts pairs of objects that share a message as maximally similar in message space. In a compressed language such collisions are semantically local, so topsim rises with compression even when the collision-free geometry does not. All structure claims in the main text therefore use topsim_distinct (pairs with distinct messages only) and CBM; plain topsim is reported only where a registered hypothesis named it (H1–H9, C1–C6).

Supplementary Note 3 (anticipated objections and where the evidence stands). 1. *Early target without loss of final alignment (K17c vs K17b).* Immature anchors redirect siblings toward the parent's step-500 partition decisively (ARI 0.43 vs 0.21, 15/15 and 19/20), but the predicted reduction in alignment with the parent's final partition reached only 70–73% of seeds with a CI above zero (Table S1). The reconstruction model illustrates one reason this need not occur: where the early and final partitions of a parent overlap (ARI 0.70 in the model), moving toward the early state need not move away from the final one; whether the neural partitions overlap to the same degree was not measured. The main text reports K17b as consistent but unconfirmed. 2. *Capacity as coverage (K12).* The additional pooled subgroup with three or more mature anchors met the equivalence bands in both cohorts. The registered subgroup (two or more) met them in confirmation but not in replication (+0.064 [+0.033, +0.094]); objects with no anchor retain the parent's form less at capacity 40 than at capacity 8 (−0.10 to −0.12, twice). The bins overlap, so no cutoff at exactly three anchors is inferred. One hypothesis, not tested here, is that at high capacity an unanchored object sits among many taught neighbours of other classes and is pulled by them, whereas at low capacity it is free to drift. The failure is reported in the main text and in Table S1; the coverage claim is scoped to well-anchored objects and stated as an association. 3. *One-shot decay vs recurring persistence (S1–S3 vs K18).* A single immature snapshot at generation 1 redirects strongly (ARI to the snapshot 0.40 vs 0.23) but the redirection is inconclusive by generation 3 under refreshed records, whereas an accumulating record re-supplies immature provenance every generation and its founder signature is measurably weaker at generation 5 (K18). The persistence claim in the main text is therefore scoped to recurring immature provenance; redirection weakens after refreshed transmission, and the longer-term persistence of a one-shot event remains inconclusive. 4. *The near-injective learner (A1–A4).* The MLP learner's languages have almost no message classes, so partition measures are uninformative by construction and class-matched anchoring has nothing to act on. Form-level tests registered on fresh seeds show that symmetry breaking (A3′) and the provenance-target shift (A4′) survive; what does not survive is the class-matched mechanism and the "coordination without loss" finding. This is presented as a regime boundary, not as a failure of the claim, because the boundary was predicted in the decision tree committed before the results were read (`results_arch/PREREG.md`).

S2. Supplementary Tables

Table S1. Core pre-registered tests on the main (GRU) learner: confirmation cohort and independent replication

Confirmation cohorts: K1–K13 and K15 on seeds 10–29 (`results_v3_confirm2`, 20 seeds, four channel families: capacity 8, capacity 40, noise 0.2, reader = both, each compared with its capacity-19 baseline where applicable); K14, K16, K17 on seeds 30–44 with parents trained fresh (15 seeds), K14 and K17b extended to seeds 30–74 (n = 45). Replication: seeds 100–119 by one command (`replicate.py`), 20 seeds. Family order in multi-family cells: capacity 8 / capacity 40 / noise 0.2 / reader both.

id	prediction (metric)	confirmation cohort	replication (seeds 100–119)	verdict
K1	capacity 8 < capacity 19 (topsim_distinct)	−0.072 [−0.089, −0.055], 20/20	−0.075 [−0.092, −0.057], 19/20	supported ×2
K2	capacity 19 = capacity 40, two-sided (topsim_distinct)	−0.017 [−0.033, −0.003]: 40 higher	−0.027 [−0.044, −0.010]: 40 higher	not supported (40 slightly higher, CI excludes 0, both cohorts)
K3	capacity 19 > 40 (CBM)	+0.024 [+0.015, +0.034], 17/20	+0.015 [+0.003, +0.027], 13/20	supported once; weakened in replication (CI > 0, seed rule not met)
K3	capacity 19 > 40 (owners-only topsim)	+0.028 [+0.005, +0.049], 14/20	+0.019 [−0.002, +0.038], 14/20	inconclusive ×2
K4	founder intelligibility 40 > 19 > 8, each step ≥ +0.15 (continuity)	+0.195 and +0.205, 20/20 each	+0.228 and +0.194, 20/20 each	supported ×2
K5	corr(continuity, topsim_distinct) > 0; corr(continuity, convexity) < 0, across 30 cells	+0.86 / −0.68	+0.88 / −0.67	supported ×2 (descriptive correlations, no CI)

id	prediction (metric)	confirmation cohort	replication (seeds 100–119)	verdict
K6	noise 0.2 < noise 0 (topsim_distinct)	−0.010 [−0.025, +0.006], 15/20	−0.013 [−0.022, −0.004], 12/20	not supported (seed rule not met ×2)
K6	noise 0.2 < noise 0 (CBM)	−0.023 [−0.030, −0.015], 17/20	−0.020 [−0.026, −0.013], 17/20	supported ×2
K7	noise 0.2 > noise 0 (number of owners)	+4.8 [+3.6, +5.9], 19/20	+4.4 [+3.4, +5.5], 19/20	supported ×2
K8	reader = both vs sender: no difference (topsim_distinct, CBM, held-out accuracy)	no difference ×2; CBM −0.008 [−0.014, −0.001]	no difference ×3	descriptive: no difference detected for geometry and held-out accuracy; CBM slightly lower in confirmation, inconclusive in replication; no equivalence band was registered, so structural equivalence is not established
K8	reader = both > sender: fidelity of taught forms	+0.052 [+0.040, +0.064], 20/20 seeds	+0.053 [+0.044, +0.061], 20/20	supported ×2 (seed as the unit, matched-cell differences averaged within seed; Version 1.0 pooled 120 dependent seed × cell pairs)
K9	accumulate > rewrite (convexity)	+0.010 [−0.052, +0.076], 9/20	−0.011 [−0.059, +0.038], 9/20	failed ×2
K9	capacity 8 > 40 (convexity)	+0.216 [+0.165, +0.266], 20/20	+0.152 [+0.095, +0.209], 17/20	supported ×2
K10	child repeats the parent's borrowed choice: taught source − untaught source ≥ 0.15; untaught within 0.10 of chance	+0.40 / +0.38 / +0.36 / +0.40, 20/20 ×4; untaught − chance −0.06 to +0.02 (chance computed on the untaught cases only)	+0.38 / +0.30 / +0.36 / +0.41, 20/20 ×4; −0.08 to +0.03	supported ×2 (family-pooled; the registered "in every cell" wording was not tested cell by cell; chance is a stipulated uniform-neighbour reference)
K11	per-seed slope of retention on number of taught neighbours > 0 (registered as monotone in every cell)	slope +0.08 / +0.12 / +0.07 / +0.09 per anchor, 20/20 ×4; adjacent steps (family-pooled per seed) weaker: capacity 40 13/20 per step, capacity 8 15/20 for 1→2 and 2→3+	slope +0.07 / +0.12 / +0.07 / +0.10, 20/20 ×4; adjacent steps 15–20/20	supported as an aggregate slope; every-step monotonicity per cell not established
K12	retention at equal anchor count, capacity 40 = capacity 8 (equivalence: |mean| < 0.05, CI within ±0.08); registered subgroups 0 vs ≥ 2	≥ 2 anchors (registered): +0.038 [+0.009, +0.068] supported; ≥ 3 anchors (additional, non-registered): −0.007 [−0.052, +0.036] supported; 0 anchors: −0.124 [−0.149, −0.097] failed	≥ 2: +0.064 [+0.033, +0.094] failed; ≥ 3: +0.023 [−0.012, +0.055] supported; 0: −0.101 [−0.116, −0.085] failed	registered subgroup equivalent once out of twice; additional 3+ subgroup equivalent twice; zero-anchor stratum failed twice (Version 1.0 reported only the 3+ subgroup as the registered test)
K13a	parent–child ARI, rewritten record − no record > 0	+0.229 / +0.643 / +0.278 / +0.426, 20/20 ×4	+0.196 / +0.610 / +0.251 / +0.394, 20/20 ×4	supported ×2
K13b	parent–child ARI, rewrite − accumulate > 0 (random / success selection)	capacity 40: +0.394 (20/20), +0.371 (19/20); noise: +0.046 (13/20), +0.103 (19/20); reader both: +0.133 (15/20), +0.108 (16/20); capacity 8: −0.024 (9/20), −0.103 (4/20)	capacity 40: +0.375 (19/20), +0.389 (20/20); noise: +0.088, +0.065 (16/20 each); reader both: +0.112 (15/20), +0.064 (13/20); capacity 8: −0.013 (9/20), −0.061 (6/20)	supported at capacity 40; reversed at capacity 8 ×2 (the coverage boundary, main text §2.4); mixed elsewhere

id	prediction (metric)	confirmation cohort	replication (seeds 100–119)	verdict
K14	content rule vs random 19: |Δtopsim_distinct| < 0.03 and |ΔCBM| < 0.02 for ≥ 3 of 4 rules (point-estimate bands; CIs not required inside the band)	seeds 30–44: 2/4 (failed); seeds 30–74 pooled: 4/4 (supported)	3/4 (whole classes: Δtopsim_distinct −0.031, outside the band)	supported at n ≥ 20 after failing at n = 15; small effects of item choice, not an equivalence test
K15	retention with ≥ 1 same-form anchor − with other-form anchors only ≥ 0.15	+0.439 / +0.303 / +0.367 / +0.414, 20/20 ×4 (same / other: 0.56/0.12, 0.57/0.26, 0.46/0.09, 0.60/0.19)	+0.425 / +0.210 / +0.362 / +0.409; 20, 19, 20, 20 of 20	supported ×2
K16	sibling ARI: same record − different records > 0 (primary); different − none > 0 (secondary)	seeds 30–44: +0.209 [+0.113, +0.304], 13/15; +0.146 [+0.064, +0.233], 12/15 (levels 0.460 / 0.251 / 0.105)	+0.170 [+0.106, +0.236], 18/20; +0.206 [+0.151, +0.263], 19/20 (levels 0.428 / 0.259 / 0.052; sibling arms trained from the saved replication parents)	supported ×2
K17a	sibling ARI, immature − mature anchors: one-sided no-material-weakening criterion (mean ≥ −0.05, CI lower ≥ −0.10; called "equivalence" in the registration)	+0.087 [−0.003, +0.174] (0.547 vs 0.460)	+0.082 [+0.015, +0.144]	criterion met ×2 (no material weakening detected; not evidence of equality)
K17b	child–parent(final) ARI, mature − immature > 0	seeds 30–44: +0.039 [−0.042, +0.126], 9/15; pooled n = 45: +0.079 [+0.039, +0.119], 33/45 (73%)	+0.074 [+0.012, +0.135], 14/20 (70%)	not confirmed: CI > 0 in the pooled 45-seed and replication analyses, not in the 15-seed confirmation alone; seed rule not met in any confirmation cohort (discovery reached 26/30)
K17c	child–parent(step 500) ARI, immature − mature > 0	+0.218 [+0.161, +0.278], 15/15 (0.428 vs 0.210)	+0.168 [+0.124, +0.211], 19/20	supported ×2
K18	ARI(gen 0, gen 5): rewrite − accumulate > 0 (a); rewrite > 0.10 and accumulate < 0.10 (b)	added to the registration while the replication was running, before inspection (registration addendum of 2026-09-04); no earlier cohort	(a) +0.092 [+0.053, +0.130], 16/20; (b) 0.167 vs 0.075	supported (one cohort, exactly at the 80% rule)

Table S2. Earlier registrations: factorial sweep (H1–H9, seeds 0–9) and first confirmation (C1–C6, seeds 10–29)

These hypotheses were written before the v3 sweep and were the source of the K-series. Metrics are those named in the registration (plain topsim where stated; see Supplementary Note 2 for why later tests use topsim_distinct).

id	prediction (metric)	seeds 0–9 (n = 10)	seeds 10–29 (n = 20)	verdict
H1	random record slots beat success-selected slots (topsim)	−0.007 [−0.013, −0.002], 3/10	not run	inconclusive

id	prediction (metric)	seeds 0–9 (n = 10)	seeds 10–29 (n = 20)	verdict
H2	a smaller record yields a more compositional language (topsim)	−0.016 [−0.024, −0.008], 8/10 in predicted direction	not run	met the rule, then **retracted**: the gain is collision inflation of plain topsim; topsim_distinct reverses it (K1)
H3	rewriting from the final language beats accumulating carved forms (topsim)	+0.039 [+0.033, +0.045], 10/10	C4: +0.071 [+0.045, +0.096], 17/20	supported ×2 (the Snapshot Effect, structural cost)
H4	noise repairs an accumulating record (topsim)	−0.039 [−0.051, −0.025], 1/10	not run	refuted
H5	a record both agents read changes held-out accuracy (two-sided)	−0.004 [−0.022, +0.014]	K8: no difference	no difference ×2
H6	every record compresses the lexicon vs no transmission (unique messages)	−16.2 [−18.2, −14.2], 10/10	not run	supported
H7	fixed random slots beat success-rewritten slots (v2 replication; topsim)	−0.014 [−0.045, +0.016], 5/10	C5: −0.020 [−0.047, +0.005], 9/11	not replicated (no difference)
H8	recording the hardest objects helps held-out accuracy (test_acc)	−0.044 [−0.065, −0.019], 1/10	not run	refuted
H9	rewritten records are more mutually intelligible (intelligibility)	+0.032 [+0.018, +0.045], 9/10	not run	supported on 10 seeds; superseded by K4/K5 (continuity), not used in the main text
C1	hard + rewrite > random + rewrite (topsim)	+0.039 [−0.002, +0.084], 5/10	+0.027 [−0.004, +0.054], 14/20	inconclusive ×2
C2	hard + rewrite > random + rewrite (test_acc)	+0.088 [+0.049, +0.130], 9/10	+0.019 [−0.035, +0.068], 12/20	not replicated
C3	hard + rewrite > success + rewrite (test_acc)	+0.104 [+0.045, +0.169], 9/10	+0.010 [−0.048, +0.071], 9/20	not replicated
C4	success + rewrite > success + accumulate (topsim)	+0.085 [+0.046, +0.132], 10/10	+0.071 [+0.045, +0.096], 17/20	supported ×2
C5	random-fixed + rewrite vs success + rewrite, two-sided (topsim)	−0.014, no difference	−0.020, no difference	no difference ×2

id	prediction (metric)	seeds 0–9 (n = 10)	seeds 10–29 (n = 20)	verdict
C6	each record cell vs the no-record generation, two-sided (topsim)	accumulate cells below (−0.043, −0.064); rewrite cells no difference; hard + rewrite above (+0.046)	random + accumulate −0.049 [−0.068, −0.031] (2/18); success + accumulate −0.035; random + rewrite +0.016 (no difference); success + rewrite +0.036; hard + accumulate +0.035; hard + rewrite +0.043	a no-record generation reaches structure comparable to record lineages: rewrite cells within a few hundredths, accumulating cells below (main text §2.1)

Table S3. Boundary experiments E, L, S, M, D (all registered before running)

id	manipulation and prediction	result	verdict
E1	entropy bonus 0.005 / 0.02 / 0.08: rewrite > accumulate in topsim_distinct at each	0.005: +0.047 [+0.026, +0.068], 21/30 (extended to n = 30 as registered); 0.02: +0.085 [+0.042, +0.131], 9/10; 0.08: +0.041 [+0.027, +0.058], 10/10	supported at 0.02 and 0.08; at 0.005 CI > 0 but 70% of seeds: weakened where the parent barely changes after capture (26% stale entries vs 35% and 41%)
E2	standardisation is imitation: child entropy at step 250 < 0.5 × generation-0 entropy	100% of seeds at all three coefficients	supported
E3	within-generation form change per 250 steps is monotone in the bonus	0.102 / 0.129 / 0.289; 0.02 − 0.005: +0.027, 9/10; 0.08 − 0.02: +0.160, 10/10	supported
E4	class-matched anchor gap ≥ 0.15 at each coefficient	+0.389 (30/30), +0.385 (10/10), +0.218 (10/10)	supported
L1	6,000-step generations: rewrite > accumulate in topsim_distinct at generation end	n = 10: +0.042 [+0.005, +0.076], 7/10 (inconclusive); extended to n = 30 as registered: +0.053 [+0.031, +0.073], 24/30	supported; stale entries at generation end 53% (vs 35% at 2,000 steps)
L2	fidelity to the record keeps eroding: step 6,000 < step 2,000	+0.158 [+0.150, +0.167], 30/30	supported
L3	class-matched anchor gap ≥ 0.15 at generation end	+0.251 [+0.221, +0.280], 30/30	supported
S1	one-shot immature snapshot at generation 1, refreshed records after: redirection toward the snapshot persists at generation 3	ARI to step-500 partition, immature − mature lineage: g1 0.397 vs 0.225; g3 +0.060 [+0.005, +0.122], 19/30	inconclusive (fades within two generations)
S2	the immature lineage is less aligned with the parent's final language at generation 5	−0.037 [−0.083, +0.004], 11/30 (0.175 vs 0.139, reversed)	not supported
S3	generation-to-generation ARI unaffected (equivalence |mean| < 0.05, CI within ±0.08)	+0.056 [+0.013, +0.096]	band missed in the favourable direction (immature lineage more coordinated)
M1	medium world (256 objects, hard distractors in 50% of rounds): class-matched anchor gap ≥ 0.15	+0.072 [+0.018, +0.117], 9/10	direction supported, magnitude below the band set on the small world
M2	sibling ARI same > different (primary), different > none (secondary)	+0.103 [+0.042, +0.164], 8/10; +0.161 [+0.106, +0.220], 10/10	supported
M3	parent–child ARI, rewrite − no record > 0	+0.288 [+0.234, +0.339], 10/10	supported
M4	rewrite > accumulate in topsim_distinct	+0.021 [−0.003, +0.045], 7/10 (stale entries 56% vs 35%)	inconclusive at n = 10

id	manipulation and prediction	result	verdict
D1	hard-distractor share 0 / 0.5 / 1.0: partition-level sibling gap (same − different) decreases monotonically	gaps 0.255 → 0.142 → 0.087; 0 − 0.5: +0.113 [−0.012, +0.234], 14/20; 0.5 − 1.0: +0.055 [−0.049, +0.157], 12/20	inconclusive (trend in the predicted direction)
D2	form-level sibling gap does not decrease (equivalence |mean| < 0.10, CI within ±0.15)	0.286 → 0.222 → 0.220; 1.0 − 0: −0.067 [−0.141, +0.003]	supported
D3	descriptive: regime change	distinct messages 38.2 / 47.3 / 44.7 of 64; owner share 0.49 / 0.64 / 0.58	the GRU does not reach injectivity even at 100% hard rounds

Table S4. Second architecture (A) and minimal reconstruction model (T)

Second architecture: MLP sender and receiver with straight-through Gumbel-softmax, no REINFORCE, no entropy bonus (Supplementary Methods, "Second architecture"). Seeds 100–119 for A1–A4 and the K10–K12 analogues; seeds 120–159 (n = 40) for the form-level tests A3′/A4′, registered after the near-injectivity of this learner was seen on seeds 100–119 and before the new seeds were run. Descriptive regime: 59.3 of 64 messages distinct; 4% of training objects have a same-form neighbour (GRU: 75%).

id	prediction	result	verdict
A1 (= K13a)	parent–child ARI, rewrite − no record > 0, four families	corrected metric: +0.165 [+0.116, +0.211] (19/20) / +0.685 [+0.646, +0.723] (20/20) / +0.398 [+0.315, +0.480] (20/20) / +0.442 [+0.364, +0.524] (20/20) (Version 1.0, singleton-zero convention: +0.152 (20/20) / +0.035 (12/20) / +0.028 (15/20) / +0.126 (18/20), "partial")	supported in all four families under the corrected metric; partition inheritance holds in the near-injective learner
A2 (= K15)	class-matched anchor gap ≥ 0.15	+0.203 (18/20) / −0.076 (7/19) / +0.015 (11/20) / +0.057 (11/20)	vanishes except at capacity 8: a regime boundary of the mechanism
K10 analogue	anchored choice, taught − untaught source	+0.238 (20/20) / +0.256 (13/16) / +0.221 (19/20) / +0.202 (19/20)	3 of 4 families
K11 analogue	retention slope on taught neighbours > 0	+0.10 / +0.15 / +0.11 / +0.10, 20/20 ×4	supported
K12 analogue	retention at equal anchor count, capacity 40 = 8	0 anchors: +0.030 [−0.002, +0.062] (equivalent); ≥ 3 anchors: +0.203 [+0.150, +0.261] (differs)	pattern reversed relative to the GRU; reported for completeness
A3 (= K16, ARI)	sibling ARI same > different; different > none	corrected metric: +0.290 [+0.058, +0.516], 12/20; +0.141 [−0.005, +0.311], 15/20 (Version 1.0, singleton-zero convention: −0.010; +0.041)	inconclusive both ways; near-singleton partitions limit what partition agreement can reveal about form coordination (see A3′)
A4 (= K17a/c, ARI)	immature anchors: coordination within band; alignment to step-500 partition higher	corrected metric: K17a −0.198 [−0.477, +0.089] (band NOT met; Version 1.0 reported +0.152 [+0.027, +0.286] under the singleton-zero convention); K17c +0.113 [+0.046, +0.187], 15/20; K17b +0.029	K17a band not met and no directional decrease established either; K17c CI > 0, seed rule not met; superseded by A4′

id	prediction	result	verdict
A3′	form-level sibling agreement: same – different > 0 (primary); different – none > 0 (secondary)	+0.232 [+0.185, +0.279], 37/40; +0.204 [+0.173, +0.239], 40/40 (levels 0.44 / 0.21 / 0.00; GRU 0.47 / 0.25 / 0.01)	supported
A4′ target	sibling agreement with the parent's step-500 forms, immature – mature > 0	+0.089 [+0.070, +0.108], 38/40	supported
A4′ strength	sibling–sibling form agreement, immature – mature: not more than 0.05 below (CI lower ≥ −0.10)	−0.064 [−0.112, −0.016] (0.374 vs 0.439) (Version 1.0 printed −0.117 [−0.181, −0.053], not the 40-seed values)	failed: immature anchors coordinate less well in this learner (main text §2.4)
H3/C4 analogue	rewrite – accumulate partition inheritance (no verdict registered)	corrected metric, K13b by ARI: capacity 40 random +0.461 [+0.311, +0.597] 19/20, success +0.420 [+0.314, +0.526] 18/20 (2 ties); noise success +0.441 [+0.351, +0.530] 20/20, random +0.264 [+0.123, +0.417] 15/20; reader both success +0.342 [+0.219, +0.465] 17/20, random +0.206 [+0.062, +0.345] 14/20; capacity 8 +0.061 / −0.025 (Version 1.0 under the singleton-zero convention: e.g. capacity 40 random +0.091 [+0.010, +0.207], "inconclusive in all families")	reported without verdict, as registered; ARI measures partition inheritance, not language structure, so this is not a structural claim
T1	model siblings: same anchors > different > none (ARI)	0.414 > 0.273 > 0.236 (sibling–parent 0.454 / 0.437 / 0.213)	reproduced (neural: 0.46 > 0.25 > 0.10)
T2	class-matched anchoring emerges from the objective	untaught object keeps its parent classmates: same-class anchor 0.47 (n = 480); other-class anchors only 0.32 (391); no anchor 0.31 (29); measured move cost at the converged child (2026-09-05): with a same-class anchored neighbour the object is already in the anchor's class in 0.44 of pairs and joining otherwise costs ΔE +1.11 on average (ΔE ≤ 0 in 0.35); with an other-class anchor 0.13, +1.85, 0.13	reproduced without being built in (neural: 0.46–0.60 / 0.09–0.26 / 0.10); the earlier verbal derivation ("below w") was incorrect and is replaced by the measured cost
T3	early-partition anchors coordinate siblings comparably but onto the early partition	sibling ARI 0.368 vs 0.414; alignment to early partition 0.397 vs 0.376, to final 0.374 vs 0.454 (early vs final partitions of the same parent: ARI 0.70)	reproduced in direction; small shift because the model's early and final partitions are close
T4	founder similarity decays more slowly under refreshed than under frozen early anchors	Version 1.0 implementation drew a fresh unrelated shallow partition every generation (not the registered frozen-record condition): rewrite 0.466, 0.314, 0.327, 0.242, 0.198; "accumulate" 0.263, 0.196, 0.194, 0.158, 0.157. Corrected implementation (design fixed before running, `results_model/T4_correction_design.md`; anchors from the founder's frozen early partition, ARI founder–early 0.70): rewrite unchanged; accumulate 0.382, 0.324, 0.376, 0.399, 0.306; g5 rewrite – accumulate −0.109 [−0.175, −0.048], rewrite higher in 5/20	NOT reproduced: implemented as registered, a frozen early record keeps the lineage closer to the founder than refreshed anchors do; the correspondence with neural K18 is withdrawn (the neural accumulating record is re-carved each generation and is not a frozen founder snapshot, so the registered model condition was not its analogue)

S3. The minimal reconstruction model

Model specification

Objects are the nodes of the Hamming graph of a 3×4 world (64 nodes; edges join objects differing in one attribute). A language is a partition P of the nodes into message classes plus surface forms (one random code per class, learner-specific unless supplied by an anchor). A learner minimises $E(P) = c \cdot \#(\text{within-class pairs}) + \lambda \cdot \#(\text{classes}) + w \cdot \#(\text{cut Hamming-1 edges})$, $c = 1$, $\lambda = 6$, $w = 1$, by greedy local search from a seeded random partition. The class-matched spillover effect is not built in. Anchors are nodes whose class label is fixed by the record (two anchors with the same recorded form share a class). Minimisers are non-unique: two unanchored learners of the same world agree at ARI 0.23 (chance 0), relabellings are free, and many alternative local merges of neighbours are equal or near-equal in cost.

Simulation results (20 seeds)

prediction	result	matches the neural system?
T1 siblings: same anchors > different > none	ARI 0.414 > 0.273 > 0.236	yes (0.46 > 0.25 > 0.10; the model's "none" floor is higher because E favours a world-generic optimum)
T2 class-matched anchoring emerges	untaught object keeps its parent classmates: same-class anchor 0.47; other-class anchors only 0.32; no anchor 0.31	yes (0.46–0.60 vs 0.09–0.26 vs 0.10); not assumed, derived from E
T3 strength vs target	anchors from the early partition: sibling ARI 0.368 vs 0.414; alignment to the early partition 0.397 vs 0.376, to the final 0.374 vs 0.454	yes in direction; the model's early/final partitions are close (ARI 0.70), so the shift is small
T4 persistence	corrected run (frozen founder early partition): ARI to founder at g5 rewrite 0.198, accumulate 0.306; −0.109 [−0.175, −0.048], rewrite higher in 5/20	no: the registered model condition does not reproduce neural K18 (Version 1.0's "reproduced" rested on an implementation that used unrelated fresh partitions)

Why T2 emerges in this regime (measured, not derived; 2026-09-05)

The move cost for an untaught node o entering a class of size m_dest from a class of size m_src is $\Delta E = c \cdot (m_{dest} - (m_{src} - 1)) + \lambda \cdot \Delta\#\text{classes} + w \cdot \Delta\#\text{cut edges}$. Version 1.0 claimed the marginal collision cost of joining an anchored neighbour's class is "below w"; with $c = w = 1$ that is false in general (joining a class of size m adds m collision pairs). Measured at the converged children of the T2 simulation: with a same-class anchored neighbour, the untaught object is already in the anchor's class in 0.44 of pairs and, when it is not, joining costs +1.11 on average ($\Delta E \le 0$ in 0.35 of pairs); with an other-class anchored neighbour, 0.13, +1.85 and 0.13. The class-matched effect therefore appears as a higher share of objects that ended in the anchor's class and a lower cost of joining, not as a guaranteed downhill move. Greedy local-search endpoints are not shown to be global minimisers. A one-dimensional chain makes the mechanism transparent: with c small and $w > \lambda/2$, a class boundary placed between two anchors of different labels is free to sit anywhere, whereas an anchor sharing a segment's label pins the segment.

Why the model separates strength and target

The objective does not privilege anchor provenance: any consistent set of fixed labels constrains the search in the same way, so provenance changes the target directly (the reconstructed partition is biased toward the partition the labels are consistent with), while coordination strength changes only modestly in this regime (0.368 vs 0.414 in the simulation), through the optimisation trajectory rather than the objective.

What the model does not capture

Surface-form turnover (forms are free per class here; in the neural system forms are re-derived with partial reuse), the structural cost of immature anchors (E has no notion of "less systematic"; the neural cost arises from learning dynamics), the exact magnitudes, and persistence: the registered T4 condition, once implemented, does not reproduce neural K18 (above). It is a model of the reconstruction step, not of learning. T1–T3 numbers are from the original run (`model.py`); the verification of 2026-09-05 reviewed the implementation but did not independently rerun them; the corrected T4 and the move-cost measurement are in `model_v2.py`.

S4. Supplementary Figure Legends

Figure 1. Historical symmetry breaking among independently trained siblings. Two children of the same parent, trained separately, become substantially more similar when they were shown the same limited set of examples: same record → siblings look alike; different records → less alike; no record → almost unrelated. (a) Partition similarity (adjusted Rand index over the 48 training objects) between two independently initialised siblings taught the same 19 (object, message) pairs, two different random 19-sets, or nothing; n = 45 seeds (30 discovery + 15 pre-registered, K16). (b) Siblings taught the same record resemble each other more than either resembles the parent. (c) The same pattern in a second learner: an MLP sender and receiver trained with straight-through Gumbel-softmax (no REINFORCE, no exploration bonus) produce nearly injective languages, so agreement is measured as the share of objects on which the two siblings use identical forms; GRU (seeds 0–29) vs Gumbel MLP (seeds 120–159; A3′, pre-registered). Error bars: s.e.m. over seeds.

Figure 2. Class-matched anchors propagate transmitted conventions. A transmitted example steadies the meanings around it, especially the ones that use its convention. (a) Share of untaught objects that keep the parent's form, by number of taught neighbours, split by whether at least one neighbour shares the object's own form (K11, K15). (b) When the parent had borrowed a neighbour's form for an untaught object, the child repeats that choice if the neighbour was taught, and at chance if not (K10). (c) Part of the benefit of a bigger record is more anchors per object: at equal anchor counts, retention is similar across capacities for well-anchored objects (three or more taught neighbours, an additional subgroup); the registered subgroup of two or more did not replicate equivalence, and unanchored objects retain less at capacity 40 (K1, K12). Error bars in all panels: s.e.m. over seeds (n = 30 in every stratum except capacity 40 with one anchor in panel c, n = 29: one seed had no eligible observation). Chance in panel b is computed on the same untaught-source cases as the bar it is compared with.

Figure 3. Developmental provenance shifts the reconstruction target. In the GRU learner, examples taken from an immature stage of the parent's language coordinate children without a detectable weakening under the pre-specified criterion, but toward the immature version (in the second learner they coordinate somewhat less well; main text §2.4). Bars: sibling–sibling similarity (coordination), similarity to the parent's final language, and similarity to the parent's step-500 snapshot, for children taught mature vs immature forms of the same 19 objects (K17a, K17c; n = 45: 30 discovery + 15 pre-registered).

Figure 4. The Snapshot Effect and its structural consequences. (a) Lineages whose record accumulates early forms build less structured languages than lineages whose record is rewritten from the parent's final language, paired by seed (H3, C4; collision-free geometry). (b) Within a generation, communication erodes what imitation installed, twice as fast when the record is immature. (c) With a very small record (capacity 8) the advantage of mature records disappears or reverses; with a large one it is strong (K13b). (d) Same objects, same parent: a child taught the mature form ends with the parent's final form 79% of the time; taught the step-500 form, 26% (per-seed means, n = 65 seeds: 30 discovery, 15 pre-registered, 20 replication). Error bars in all panels: s.e.m. over seeds.

S5. Registration timeline and reproduction

File paths and commands in this section identify materials within the research code repository. They are provided for reproducibility, rather than as links to a reader's local computer. The archive linked on the title page is the original Version 1 release; the corrected analysis requires the revised code described below. The accompanying *Version 2 Revision Notes* summarize the analytical changes.

Two kinds of registration are distinguished throughout. *Registered before the runs*: H1–H9 (`results_v3/PREREG.md`), C1–C6 (`results_v3_confirm/PREREG.md`), K1–K9 (`results_v3_confirm2/PREREG.md`), K14, K16, K17 (new seeds, same file; K18 was added to that file while the replication was running, before inspection), E1–E4 (`results_entropy/PREREG.md`), L1–L3 (`results_long/PREREG.md`), S1–S3 (`results_oneshot/PREREG.md`), M1–M4 (`results_medium/PREREG.md`), D1–D2 (`results_degeneracy/PREREG.md`), A1–A4 with the A3′/A4′ addendum and its power extension (`results_arch/PREREG.md`), and T1–T4 (`results_model/PREREG.md`). *Registered before computation on held-back data*: K10–K13 and K15, quantities discovered post hoc on seeds 0–29 of the capacity-19 cells and fixed in writing before being computed on the confirmation-2 families (capacity 8 / 40, noise, reader), which had not been used for those quantities. The theory was frozen on 2026-09-04; every experiment after that date is a robustness or boundary test and none was added to the core claims. Thresholds for K1–K17 are frozen in `manifest_k.json`; a K18 entry was added on 2026-09-05 as missing metadata, with its provenance.

Reproduction. `python3 tests.py` (31 checks) before any run. `python3 replicate.py --seeds 100..119 --out DIR` is the tested workflow (see Compute for its limits); `--arch gumbel` does the same for the second architecture. `python3 model.py` reproduces the original T1–T3 and the superseded T4 implementation (kept for the record); `python3 model_v2.py` reproduces the corrected T4 and the T2 move-cost measurement. `python3 figures.py` regenerates Figures 1–4 into `figs_v2/`. Cached-data re-scoring with the corrected metric: `corrections_v2/rerun_all.sh`. CPU only; a 6-

generation run takes about two minutes on a 10-core laptop. `PROBES_INDEX.md` maps every script to its output file.

S6. References (all works cited in the main text or this Supplement)

Acerbi, A. (2021). Culture without copying or selection. Evolutionary Human Sciences, 3, e50.

Alemohammad, S., et al. (2024). Self-consuming generative models go MAD. ICLR.

Baronchelli, A., Felici, M., Loreto, V., Caglioti, E., & Steels, L. (2006). Sharp transition towards shared vocabularies in multi-agent systems. J. Stat. Mech., P06014.

Ben Zion, R., Carmeli, B., Paradise, O., & Belinkov, Y. (2024). Semantics and spatiality of emergent communication. NeurIPS (arXiv:2411.10173).

Carlsson, E., Dubhashi, D., & Regier, T. (2024). Cultural evolution via iterated learning and communication explains efficient color naming systems. J. Language Evolution, 9, 49–66.

Carmeli, B., Belinkov, Y., & Meir, R. (2024). Concept-best-matching: evaluating compositionality in emergent communication. Findings of ACL (arXiv:2403.14705).

Carr, J. W., Smith, K., Cornish, H., & Kirby, S. (2017). The cultural evolution of structured languages in an open-ended, continuous world. Cognitive Science, 41, 892–923.

Carr, J. W., Smith, K., Culbertson, J., & Kirby, S. (2020). Simplicity and informativeness in semantic category systems. Cognition, 202, 104289.

Chaabouni, R., Kharitonov, E., Dupoux, E., & Baroni, M. (2019). Anti-efficient encoding in emergent communication. NeurIPS.

Chaabouni, R., et al. (2021). Communicating artificial neural networks develop efficient color-naming systems. PNAS, 118.

Cho, J. H., & Hariharan, B. (2019). On the efficacy of knowledge distillation. ICCV.

Claidière, N., Scott-Phillips, T. C., & Sperber, D. (2014). How Darwinian is cultural evolution? Phil. Trans. R. Soc. B, 369, 20130368.

Fehér, O., Wang, H., Saar, S., Mitra, P. P., & Tchernichovski, O. (2009). De novo establishment of wild-type song culture in the zebra finch. Nature, 459, 564–568.

Griffiths, T. L., & Kalish, M. L. (2007). Language evolution by iterated learning with Bayesian agents. Cognitive Science, 31, 441–480.

Guo, S., Ren, Y., Mathewson, K., Kirby, S., Albrecht, S. V., & Smith, K. (2022). Expressivity of emergent languages is a trade-off between contextual complexity and unpredictability. ICLR.

Havrylov, S., & Titov, I. (2017). Emergence of language with multi-agent games: learning to communicate with sequences of symbols. NeurIPS.

Hubert, L., & Arabie, P. (1985). Comparing partitions. J. Classification, 2, 193–218.

Hudson Kam, C. L., & Newport, E. L. (2005). Regularizing unpredictable variation. Language Learning and Development, 1, 151–195.

Hudson Kam, C. L., & Newport, E. L. (2009). Getting it right by getting it wrong. Cognitive Psychology, 59, 30–65.

Kalish, M. L., Griffiths, T. L., & Lewandowsky, S. (2007). Iterated learning: intergenerational knowledge transmission reveals inductive biases. Psychonomic Bulletin & Review, 14, 288–294.

Kharitonov, E., Chaabouni, R., Bouchacourt, D., & Baroni, M. (2020). Entropy minimization in emergent languages. ICML.

Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory. PNAS, 105, 10681–10686.

Li, F., & Bowling, M. (2019). Ease-of-teaching and language structure from emergent communication. NeurIPS.

Lu, Y., Singhal, S., Strub, F., Courville, A., & Pietquin, O. (2020). Countering language drift with seeded iterated learning. ICML.

Panigrahi, A., Liu, B., Malladi, S., Risteski, A., & Goel, S. (2025). Progressive distillation induces an implicit curriculum. ICLR.

Perfors, A., & Navarro, D. J. (2014). Language evolution can be shaped by the structure of the world. Cognitive Science, 38, 775–793.

Puglisi, A., Baronchelli, A., & Loreto, V. (2008). Cultural route to the emergence of linguistic categories. PNAS, 105, 7936–7940.

Ren, Y., Guo, S., Labeau, M., Cohen, S. B., & Kirby, S. (2020). Compositional languages emerge in a neural iterated learning model. ICLR.

Scott-Phillips, T. C. (2017). A (simple) experimental demonstration that cultural evolution is not replicative, but reconstructive. J. Cognition and Culture, 17, 1–11.

Senghas, A., & Coppola, M. (2001). Children creating language. Psychological Science, 12, 323–328.

Senghas, A., Kita, S., & Özyürek, A. (2004). Children creating core properties of language. Science, 305, 1779–1782.

Sevestre, A., & Dupoux, E. (2025). Frequency and compositionality in emergent communication. EMNLP.

Shumailov, I., et al. (2024). AI models collapse when trained on recursively generated data. Nature, 631, 755–759.

Silvey, C., Kirby, S., & Smith, K. (2019). Communication increases category structure and alignment only when combined with cultural transmission. J. Memory and Language, 109, 104051.

Singleton, J. L., & Newport, E. L. (2004). When learners surpass their models. Language, 80, 370–407.

Skyrms, B. (2010). Signals: Evolution, Learning, and Information. Oxford University Press.

Strachan, J. W. A., et al. (2021). Evaluating the relative contributions of copying and reconstruction processes in cultural transmission episodes. PLoS ONE.

Talebirad, Y., Redman, E., Parsaee, A., & Zaïane, O. R. (2026). From signals to structure: how memory architecture drives language emergence in LLM agents. arXiv:2607.00233.

Wang, C., Yang, Q., Huang, R., Song, S., & Huang, G. (2022). Efficient knowledge distillation from model checkpoints. NeurIPS.

Xu, J., Dowman, M., & Griffiths, T. L. (2013). Cultural transmission results in convergence towards colour term universals. Proc. R. Soc. B, 280, 20123073.